

# Naive Bayes: The Intuition

## Contents

|                                                          |   |
|----------------------------------------------------------|---|
| 1. Overview . . . . .                                    | 2 |
| 2. The data . . . . .                                    | 2 |
| 3. The strategy: compare posteriors . . . . .            | 3 |
| 4. Applying Bayes' theorem . . . . .                     | 4 |
| 5. The problem: specific combinations are rare . . . . . | 4 |
| 6. The naive assumption . . . . .                        | 4 |
| 7. Computing the scores . . . . .                        | 5 |
| 8. Summary . . . . .                                     | 6 |
| 9. Key terms . . . . .                                   | 6 |

**Where this fits:** Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note 3](#)); Probability density function (PDF) ([Note 20](#)); Independent and mutually exclusive events ([Note 84](#)); Bayes' theorem ([Note 86](#)).

## 1. Overview

**Key point:** Naive Bayes computes, for each class, the probability of that class given the inputs, using Bayes' theorem, and predicts the class with the largest one. To make this possible, it assumes the inputs are independent within each class.

With conditional probability, independence and Bayes' theorem in place, we can build the **Naive Bayes classifier**. This Note develops the intuition on a tiny example; the next derives the mathematics in general, and the one after codes it.

## 2. The data

**Key point:** Eight cricket matches, three inputs (toss, venue, outlook) and the result. Predict the result of a new match.

The example is a small made-up dataset about one cricket team. Each row is a match, with three inputs and one output:

| match | toss | venue   | outlook  | result |
|-------|------|---------|----------|--------|
| 1     | won  | Chennai | sunny    | win    |
| 2     | won  | Mumbai  | sunny    | win    |
| 3     | lost | Chennai | sunny    | win    |
| 4     | won  | Mumbai  | overcast | win    |
| 5     | won  | Chennai | sunny    | win    |
| 6     | lost | Mumbai  | sunny    | loss   |
| 7     | lost | Chennai | rainy    | loss   |
| 8     | won  | Mumbai  | rainy    | loss   |
| new   | lost | Mumbai  | sunny    | ?      |

**Figure 1:** Eight matches and a new one to predict

- **toss:** did the team win or lose the toss?
- **venue:** Chennai or Mumbai.
- **outlook:** the weather, sunny, overcast or rainy.
- **result:** win or loss, the class to predict.

For a new match the toss is **lost**, the venue is **Mumbai** and the weather is **sunny**. Will the team win?

### 3. The strategy: compare posteriors

**Key point:** Compute  $P(\text{win} \mid \text{inputs})$  and  $P(\text{loss} \mid \text{inputs})$ ; predict whichever is larger.

Naive Bayes asks two questions:

$$P(\text{win} \mid \text{lost, Mumbai, sunny}) \quad \text{and} \quad P(\text{loss} \mid \text{lost, Mumbai, sunny})$$

The commas mean “and”: the three conditions hold together, so they could also be written with  $\cap$ . If the first probability is, say, 0.56 and the second 0.27, it predicts a win. With more classes (for example win, loss, draw) it computes one probability per class and again takes the largest.

## 4. Applying Bayes' theorem

**Key point:** Each posterior equals likelihood  $\times$  prior / evidence. The evidence is the same for every class, so it can be dropped.

With  $A$  = the class and  $B$  = the three input values, Bayes' theorem gives

$$P(\text{win} \mid \text{lost, Mumbai, sunny}) = \frac{P(\text{lost, Mumbai, sunny} \mid \text{win}) \times P(\text{win})}{P(\text{lost, Mumbai, sunny})}$$

and the same with “loss” in place of “win”.

The denominator, the evidence, is identical in both: it is just the probability of seeing these inputs, whichever the class. Dividing two numbers by the same positive amount does not change which is larger. So for choosing the class, only the **numerator** matters:

$$\text{score}(\text{class}) = P(\text{inputs} \mid \text{class}) \times P(\text{class})$$

The **priors** come straight from counting: 5 of the 8 matches were wins and 3 were losses.

$$P(\text{win}) = 5/8 \quad P(\text{loss}) = 3/8$$

## 5. The problem: specific combinations are rare

**Key point:** No winning match had exactly (lost, Mumbai, sunny), so the likelihood  $P(\text{lost, Mumbai, sunny} \mid \text{win})$  would be 0.

The likelihood  $P(\text{lost, Mumbai, sunny} \mid \text{win})$  asks: among the 5 winning matches, how many had all three conditions at once? None did. So it would be 0, and the score for “win” would be 0 however strongly the other evidence pointed to a win.

This is a general problem. A row matching a new input on **every** column at once is rare. With 3 columns it already failed here; with 10 or 20 columns, almost every combination would never have been seen. The estimates would be 0 or based on one or two rows, and the classifier would be useless.

## 6. The naive assumption

**Key point:** Assume the inputs are independent within each class. Then the joint likelihood becomes a product of one-column likelihoods, each estimated from many rows.

Naive Bayes makes a simplifying assumption: **within each class, the inputs are independent** of each other. From the independent events Note, the probability of independent events happening together is the product of their probabilities. So

$$P(\text{lost, Mumbai, sunny} \mid \text{win}) \approx P(\text{lost} \mid \text{win}) \times P(\text{Mumbai} \mid \text{win}) \times P(\text{sunny} \mid \text{win})$$

Each factor looks at one column only, so it is estimated from all the rows of that class. The assumption is rarely exactly true, which is why the method is called **naïve**, but it makes the estimates workable. The next Note derives this step properly.

## 7. Computing the scores

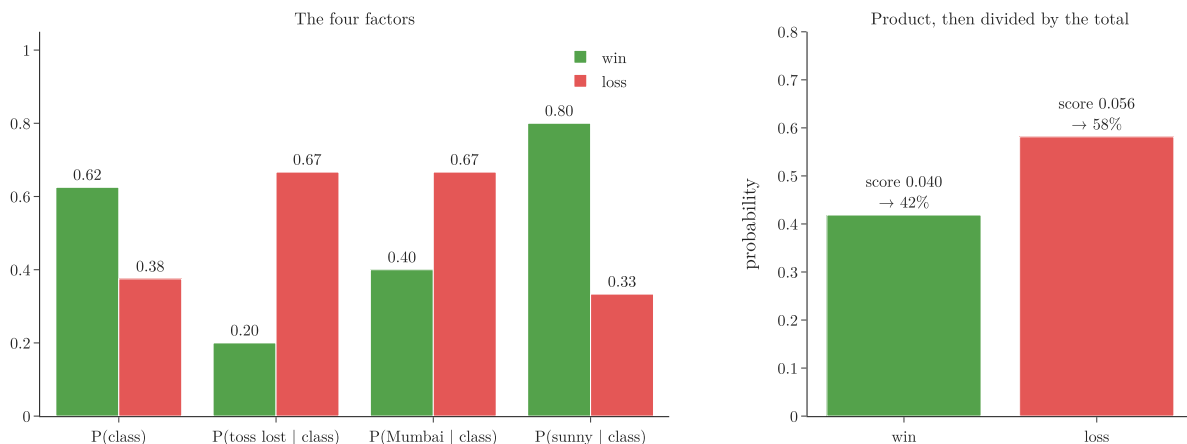
**Key point:** Win:  $5/8 \times 1/5 \times 2/5 \times 4/5 = 0.040$ . Loss:  $3/8 \times 2/3 \times 2/3 \times 1/3 = 0.056$ . Predict loss.

**Win** (5 matches): the toss was lost in 1 of them, the venue was Mumbai in 2, and the weather was sunny in 4.

$$\text{score}(\text{win}) = \frac{5}{8} \times \frac{1}{5} \times \frac{2}{5} \times \frac{4}{5} = 0.040$$

**Loss** (3 matches): the toss was lost in 2, Mumbai in 2, sunny in 1.

$$\text{score}(\text{loss}) = \frac{3}{8} \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = 0.056$$



**Figure 2:** The four factors for each class, and the resulting probabilities

The loss score is larger, so Naive Bayes predicts a **loss**. Dividing each score by their sum turns them into probabilities:  $0.040/0.096 = 42\%$  win and  $58\%$  loss (Figure 2, right).

Figure 2 (left) shows why: winning teams mostly won the toss and played in sunny weather, so “lost the toss” (0.20 against 0.67) and “Mumbai” (0.40 against 0.67) both point to a loss. Sunny weather points to a win (0.80 against 0.33), but not strongly enough to outweigh them.

## 8. Summary

| Step                                 | Win   | Loss         |
|--------------------------------------|-------|--------------|
| Prior $P(\text{class})$              | 5/8   | 3/8          |
| $P(\text{lost} \mid \text{class})$   | 1/5   | 2/3          |
| $P(\text{Mumbai} \mid \text{class})$ | 2/5   | 2/3          |
| $P(\text{sunny} \mid \text{class})$  | 4/5   | 1/3          |
| Score (product)                      | 0.040 | <b>0.056</b> |

- Naive Bayes picks the class with the largest  $P(\text{class} \mid \text{inputs})$ .
- The evidence is shared by all classes, so only likelihood  $\times$  prior is compared.
- Exact combinations of inputs are rare; the naive independence assumption splits them into one-column probabilities.

## 9. Key terms

| Term                   | Meaning                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------|
| Naive Bayes classifier | A classifier that applies Bayes' theorem with the assumption that inputs are independent within each class |
| Naive assumption       | The assumption that the inputs are conditionally independent given the class                               |
| Score                  | Likelihood $\times$ prior for a class; proportional to the posterior                                       |
| Class prior            | The share of training rows in a class                                                                      |